

Chapter 2

The Entrepreneurial Ecosystem and Global Prosperity



Introduction

While a focus on the entrepreneurial ecosystem may seem a novel approach to development, it is consistent with and even complementary to older, more traditional development strategies. As developing economies move from centralized economies to market economies, enterprise and entrepreneurship become important. “The emerging world, long a source of cheap labor, now rivals developed countries for business innovation. Developing countries are becoming hotbeds of business innovation in much the same way as Japan did from the 1950s onwards.”¹

Entrepreneurship is considered an important mechanism that promotes economic development through employment, innovation, and welfare, but it does not appear like manna from heaven as a country moves through the stages of development. Rather, it plays a role in all development stages and is a process that continues over many years. Economists have come to recognize the “input-competing” and “gap-filling” capacities of entrepreneurial activity in development.² In other words, someone has to create the technology for new products and create the markets where people will buy them.

Two points are important when thinking about entrepreneurship and development. *First, contrary to popular belief, the most entrepreneurial countries in the world are not those that have the most entrepreneurs.* This notion is in fact misleading. In fact, the highest self-employment rates are in low-income countries such as Zambia and Nigeria. This is because low-income economies lack the human capital and infrastructure needed to create high-quality jobs. The result is that many people sell soft drinks and fruit on street corners, but there are few innovative, high-growth startups. Nor do these street vendors represent business ownership as defined in many developed countries.

¹ Woolridge (2009).

² Leibenstein (1968).

In entrepreneurship, quality matters more than quantity. To be entrepreneurial, a country needs to have the best entrepreneurs, not necessarily the most. What the “best and the brightest” do is important, and to support their efforts, a country needs a well-functioning entrepreneurial ecosystem (watch the video).³ The path to development is to create efficient organizations able to harness technology to increase output and improve the lives of millions.

Second, entrepreneurship comes in productive, unproductive, and destructive forms. While productive entrepreneurship makes both entrepreneurs and society better off, unproductive and destructive entrepreneurship makes entrepreneurs better off but leave society in worse condition. The GEI strives to measure only productive entrepreneurship that both creates wealth and is scalable.

Entrepreneurial ecosystems support innovative, productive, and rapidly growing new ventures. They consist of multiple interactive elements, all of which need to be in sync in order for innovative and high-growth firms to prosper. Such firms also need skilled employees. They need access to technology. They need a well-functioning infrastructure. They need specialized advice and support. They need access to finance. They need business premises. They need a supportive regulatory framework.

The Global State of Entrepreneurship

The GEI measures both the quality of entrepreneurship in a country and the extent and depth of the supporting entrepreneurial ecosystem. The map below presents a snapshot of the global entrepreneurial ecosystem (Fig. 2.1).

The top ten countries for 2018 show a pattern similar to last year’s—high-income, mostly European nations. The top countries are the United States, Switzerland, Canada, Sweden, Denmark, Iceland, Australia, United Kingdom,

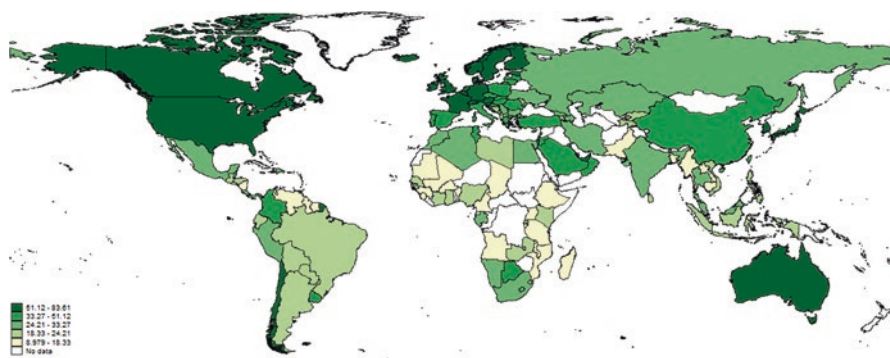


Fig. 2.1 Global Entrepreneurship 2018 Index Map

³ https://www.youtube.com/watch?v=hjNc_BScn-s.

Ireland, and the Netherlands. The major surprise this year is the movement of Switzerland from eighth place to second; the United States remains first for the third year in a row. Because the scores in the highest range are so close, small changes in score from 1 year to the next can produce a relatively large shift in ranks among the top ten. For this reason, we present confidence intervals for the top ten.

Top Ten Countries (Table 2.1 and Fig. 2.2)

The results show that the No. 1 rank could have gone to any of the top eight nations with the exception of the United Kingdom and the Netherlands. We see that Switzerland has a confidence interval almost similar to the United States.

Regional Performance

For many countries, a regional benchmark is more relevant for identifying best practices for fostering entrepreneurship. This year we have several important changes in Europe, sub-Saharan Africa, and the MENA countries. Below we present the top performer in each region along with individual and institutional score summaries (Table 2.2).

The United States leads the world in entrepreneurship and is first in the North American region, just ahead of peer Canada. Australia ranks first in the Asia-Pacific region, ahead of economic powerhouses China, Singapore, Hong Kong, and Japan.

Table 2.1 Top ten countries in the GEI

Country	GEI 2018 lower limit	GEI 2018 upper limit	GEI 2018	Rank 2018	GEI Rank 2017	Rank 2017
United States	77.5	89.7	83.6	1	1	United States
Switzerland	72.5	88.4	80.4	2	2	Switzerland
Canada	73.9	84.6	79.2	3	3	Canada
United Kingdom	73.6	81.9	77.8	4	8	Sweden
Australia	69.0	82.0	75.5	5	7	Denmark
Denmark	64.8	83.8	74.3	6	5	Iceland
Iceland	63.6	84.7	74.2	7	6	Australia
Ireland	66.8	80.6	73.7	8	9	United Kingdom
Sweden	67.1	79.1	73.1	9	4	Ireland
France	59.9	77.1	68.5	10	13	Netherlands

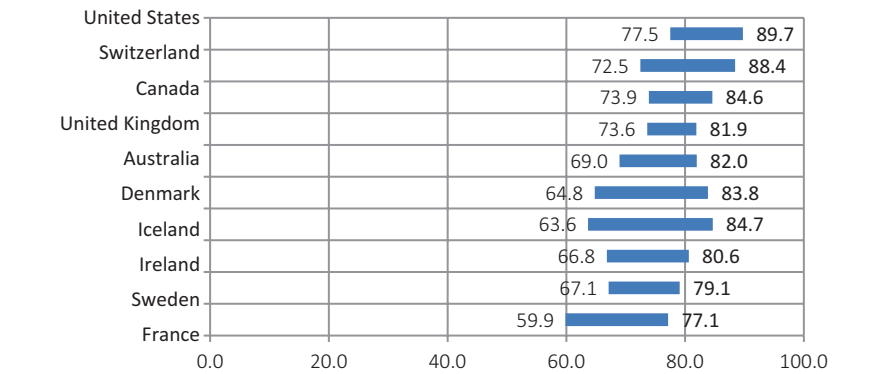


Fig. 2.2 Confidence intervals for top ten scores

Table 2.2 Top scores by region

World rank	Country	Region	GDP per capita PPP	Individual variables	Institutional variables	GEI
1	United States	North America	\$52.676	93.0	78.5	83.6
2	Switzerland	Europe	\$54.933	93.8	71.5	80.4
5	Australia	Asia-Pacific	\$42.149	82.2	74.3	75.5
16	Israel	Middle East/North Africa	\$31.092	80.9	72.7	65.4
19	Chile	South and Central America/Caribbean	\$21.302	68.9	75.9	58.5
52	Botswana	Sub-Saharan Africa	\$15.286	47.7	66.0	34.9

Switzerland, which ranked fourth in the European region and eighth overall last year, now comes in first in Europe. Chile ranks first in South and Central America and the Caribbean (16th overall), 19 places ahead of the next highest scorer in the region—Puerto Rico, at 35th. Israel is 17th overall and tops the MENA region, just ahead of UAE at 21st. In sub-Saharan Africa, Botswana is the leader at 52nd, ranking ahead of nine European nations.

Biggest Gains (Table 2.3)

Table 2.3 Biggest gains in GEI score

Country	Score 2018	Score 2017	Difference in score	Difference in rank
United Kingdom	77.8	71.2	6.5	4
Israel	65.4	59.4	6.0	1
Bulgaria	27.8	22.7	5.1	13
China	41.1	36.1	5.0	5
Iran	26.8	22.1	4.7	13
Italy	41.4	37.1	4.3	4
Poland	50.4	46.5	3.9	1
Canada	79.2	75.5	3.7	0
Ireland	73.7	70.2	3.5	1
Korea	54.2	50.7	3.5	3

Legend: includes only those countries that have participated in the GEM survey and do not have estimated individual data

Biggest Declines (Table 2.4)

Table 2.4 Biggest declines in GEI score (only with decreasing GEI scores)

Country	Score 2018	Score 2017	Difference in score	Difference in rank
Latvia	40.5	43.0	−2.5	−6
Sweden	73.1	75.2	−2.1	−5
Taiwan	59.5	61.0	−1.5	−2
Malaysia	32.7	33.4	−0.7	−4
Ecuador	20.5	20.9	−0.4	−5
Chile	58.5	58.9	−0.4	−1
Kazakhstan	29.7	30.0	−0.3	−2
Indonesia	21.0	21.1	−0.1	−4

Legend: includes only those countries that have participated in the GEM survey and do not have estimated individual data

Country-Level Productivity and the Entrepreneurial Ecosystem

Perhaps the first point that should be made in this subchapter is that economic growth does not equal to productivity. Economic growth basically refers to the capacity of countries to produce more goods and services, irrespective of how higher production is achieved. The positive variations in GDP or employment over

time are the usual suspects among those interested in studying economic growth figures, mostly because they represent the desired objective of most policymakers, as a measure of economic prosperity.

Productivity is a more complex concept. At the country level, total factor productivity (TFP) deals with two highly interconnected economic aspects. First, TFP has to do with the capacity of countries to allocate and exploit available resources efficiently (P = productivity effect). The notion that markets are good at directing resources is a good catch-all explanation concept; but for many businesses, it is hard to find all that is required to perform in the market and to keep the pace of industrial and digital revolutions that not only equip businesses with new—often more technologically advanced—resources but also change the ways to exploit them.

The second component of TFP deal with the capacity of organizations to channel innovations to the economy (I = innovation effect) that, consequently, translate into higher levels of output per input unit (in the case of countries, GDP per worker). Maybe we all are too used to link innovation to technological inventions that are successfully commercialized. However, our definition of innovation is not restricted to engineering (such as the driverless car) or to medical advances (such as nerve stimulation or noninvasive procedures) and is open to other, equally valuable, types of non-technological innovations related to product and processes.

Let's start with the productivity effect (P). The efficient allocation of resources available in the economy is an important part of the productivity function. The productivity effect is linked to how well new and existing businesses use different resources, including labor, capital, equipment, knowledge, and technology-based inputs. The capacity of Amazon to amalgamate technologies brought from other industries (e.g., ICTs, drones) to increase the productivity of its operations (delivery: AmazonFresh or Amazon Prime Air) is a good example.⁴

From the perspective of the entrepreneurship ecosystem, better institutions backing entrepreneurial activities and an efficient interaction between individual actions and the institutional setting governing entrepreneurial decisions are key ingredients necessary to facilitate the creation of businesses with a greater capacity to generate jobs and help incumbent businesses to take advantage of better market conditions. For example, in many European countries, entrepreneurs have strong incentives to invest in physical and human capital and to promote the exploitation of resources in an effort to improve the functioning of their businesses. In this case, the supportive institutional environment creates the conditions to promote operational improvements. In other words, entrepreneurial ventures have incentives to “do things better,” that is, to improve their productivity. To sum up, a healthy entrepreneurial ecosystem contributes to national productivity by enhancing market efficiency levels and by promoting the efficient exploitation of resources through new and incumbent businesses.⁵

⁴The Wall Street Journal (2014). ‘Amazon Reports Loss Despite Strong Sales Growth’. Available at: <http://online.wsj.com/articles/amazon-reports-another-loss-despite-strong-sales-growth-1406233529>.

⁵Acs, Z.J., Autio, E., Szerb, L. (2014). National systems of entrepreneurship: Measurement issues and policy implications. *Research Policy*, 43(3), 476–494.

The second effect—innovation (*I*)—is strictly linked to the Schumpeterian approach to entrepreneurship (creative destruction).⁶ For Schumpeter, entrepreneurs play a decisive role in the economy by creating and implementing radical innovations that are conducive to economic progress. In this tradition entrepreneurship is critical to spark economic development by promoting innovations, in our terminology “create new things or find new ways to do things.” Progress translates in the expansion of the countries’ production possibilities that materialize in a shift of the global frontier.

But, at this point is worth questioning how can radical innovations foster such progress. Moreover, how does the entrepreneurial ecosystem contribute to this progress? It seems logical to argue that inventions are worthless if they do not turn into commercialized innovations and that the economic impact of such innovations will turn sterile if the market and individuals cannot fully incorporate these innovations in their day-to-day routines.

For Schumpeter, entrepreneurs nurture the economy with innovations and the entrepreneurial ecosystem is critical for the development of this economic function: “create new things or find new ways to do things.” If countries enjoy a healthy entrepreneurial ecosystem, the efforts of innovative entrepreneurs will materialize in new value-adding combinations of resources that will expand the countries’ productive capacity and the global production frontier.

We found a significant, relatively strong positive correlation between entrepreneurship and total factor productivity (0.35). We also noted that entrepreneurship correlates weakly positively with the productivity effect (0.09). The strongest positive correlation was found between entrepreneurship and the innovation effect (0.39). This result is not surprising if we think a little harder. Just like we cannot imagine progress in the nineteenth century without the creation and development of steam engines, it is hard to imagine entrepreneurship in the twenty-first century without the power of technology-driven inventions. With the new millennium, industries and markets from all around the globe are witnessing drastic transformations that are the result of a digital revolution in which entrepreneurs are taking an active role by creating new businesses that are responsible of this revolution. The result is a good sign that reinforces our argument that the creation of “new things or new ways to do things” definitely constitutes the vital force driving economic development.

We also examined the connection between the GEI score and the computed total factor productivity values. The correlation between TFP and GEI is 0.35 and the sign is positive: the quality of the entrepreneurial ecosystem (GEI scores) and TFP moves in the same direction. Countries with a low-quality entrepreneurial ecosystem tend to show negative TFP values below unity. On contrary, all developed economies with supportive entrepreneurial ecosystems improve their total factor productivity, either by productivity or innovation effects.

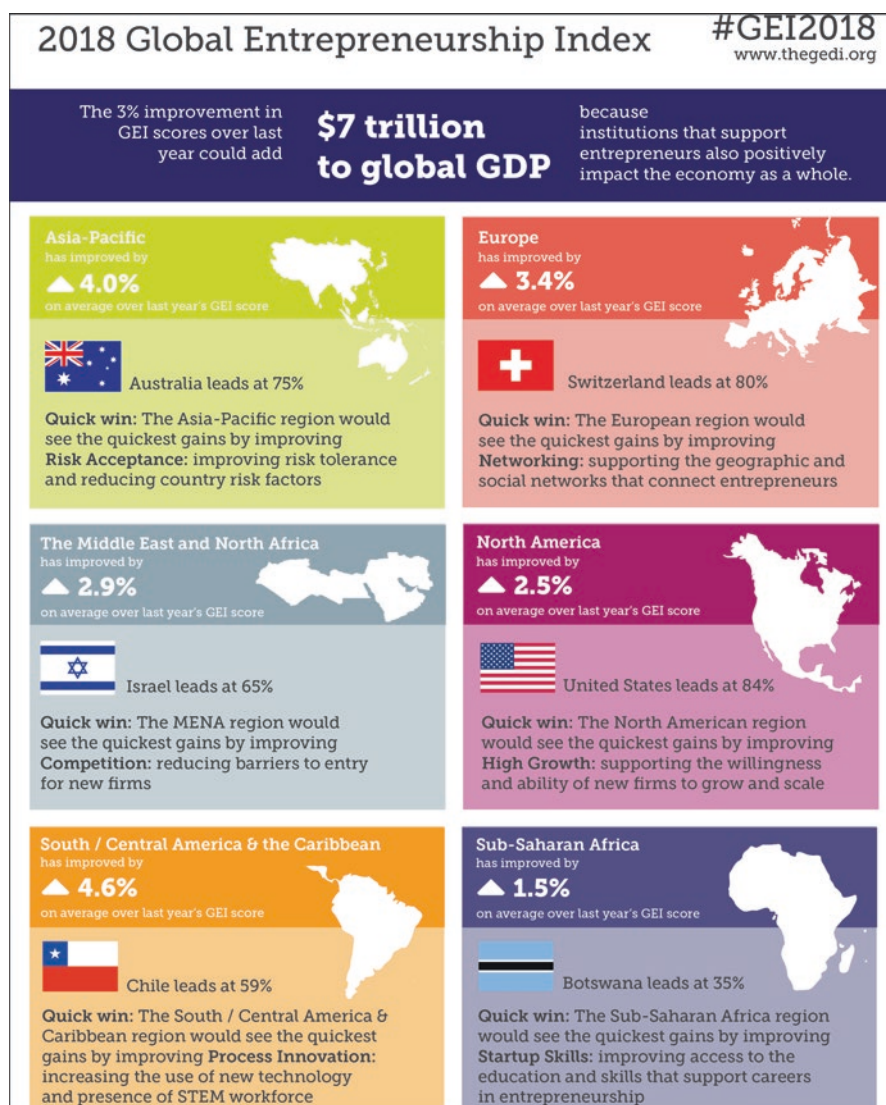
⁶Schumpeter, J.A. (1934). *The Theory of Economic Development: An Inquiry into Profits, Capital, Credit, Interest, and the Business Cycle*. Cambridge, MA: Harvard University Press.

As noted earlier, there is a positive association between entrepreneurship and the productivity effect (correlation = 0.09); however, this relationship is less pronounced than that found for the TFP. This result may well be partly explained by the differentiating impact of entrepreneurship over the productivity effect across economies. We observe that the correlation between entrepreneurship and the productivity effect scores the highest among factor-driven countries (0.47). We also note that in many underdeveloped and developing territories with low- and mid-level entrepreneurial ecosystem, the productivity effect is positive, while the result of the productivity effect for some developed economies is negative. Thus, our results suggest that, in developing economies, the entrepreneurial ecosystem plays a much more decisive role on TFP via productivity improvements, that is, helping new and established businesses in developing economies to better exploit their limited resources, that is, “to do things better.”

The picture is quite different when we look at the results for the innovation effect. The correlation between entrepreneurship and the innovation effect progressively increases as we move from factor-driven (correlation = -0.41) to innovation-driven economies (correlation = 0.33). Similarly, the impact of the innovation effect is much more potent in innovation-driven economies (1.55%) than in efficiency-driven (1.17%) and in factor-driven economies (-0.59%). In contrast to the stronger effect of the entrepreneurial ecosystem over productivity in developing countries, we found that the positive influence of a healthy entrepreneurial ecosystem over the innovative capacity of new and established businesses is much more powerful in developed economies. With the exception of Turkey, the innovation effect linked to the efficient commercialization of innovations and new technologies is positive in all economies with high-quality entrepreneurial ecosystems ($GEI > 50$).

Based on the relationships reported above, the improvement of the entrepreneurial ecosystem (GEI scores) may well yield to improvements in total factor productivity via the enhanced capacity of businesses to use their available resources (productivity effect) and to exploit the market potential of new technologies and innovations. As a result, if every of the 64 analyzed countries raised its GEI score by 10%, the global total factor productivity will increase 0.22 TFP points, which represents an estimated improvement of 15.80%.⁷

⁷We agree that correlation and basic regression coefficients are not enough to determine causation, which is why we say that a 10% global increase in GEI could improve global productivity 0.22 points. But, it should be kept in mind that the said change in the GEI, and the changes to institutions that it represents, could add this amount to global productivity, since the institutions that support entrepreneurship also support a variety of other economic and non-economic activities.



References

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